

SWT41051: Practical for Advanced Software Engineering
Department of ICT
Faculty of Technology
South Eastern University of Sri Lanka

Title: Designing a Calculator App using Visual Studio and Controllers

Aims:

- To design and develop a functional application by creating an interactive user interface and implementing event-driven programming through controllers.
- To understand form design, control properties, and button event handling to perform basic arithmetic operations, along with clear and decimal point functionality.

Exercise 01:

Design a mini calculator as shown in the figure given below to perform the basic mathematical operations.

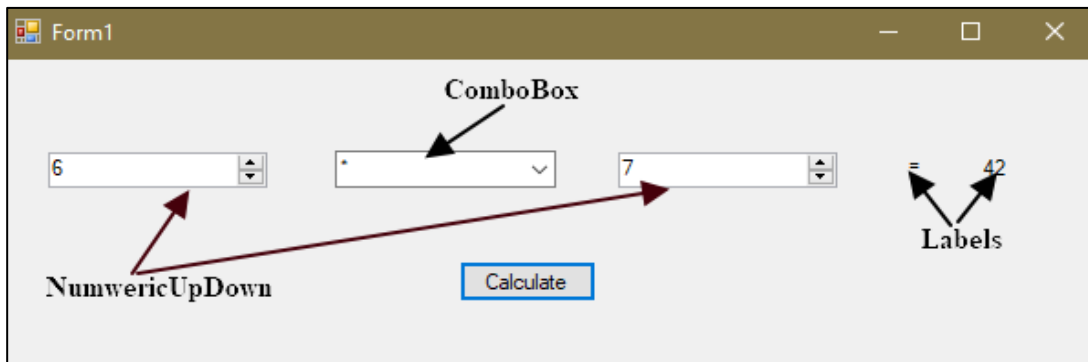


Figure 1: Mini Calculator UI Design

Exercise 02:

Design a calculator app using the following instructions given below:

1. Open your Visual Studio or Visual C# Express Edition and create a new project. set its name to **CalculatorApp**.
2. Set the following properties for Form
 - a. Text – Calculator
 - b. FormBorderStyle - FixedSingle
 - c. MaximizeBox – False
 - d. BackColour – As you wish
3. Add a TextBox controller into the form and set the properties as following:
 - a. BackColor - ControlLightLight
 - b. Font – Cambria, 16pt
 - c. ForeColor - WindowText
 - d. Multiline – False
 - e. ReadOnly – True
 - f. Text – 0 (Zero)
 - g. TextAlign – Right
4. Add the buttons as given below: Give the names of each button as given and the Text as in the figure below.

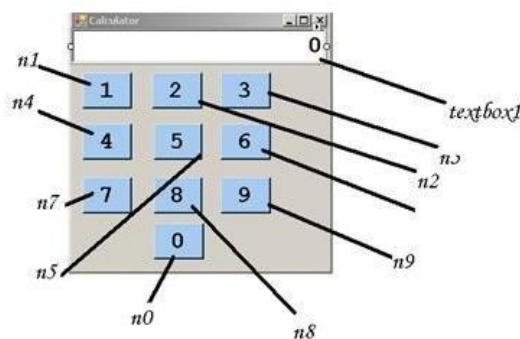


Figure 2: Calculator App

Set the **Backcolour** of the button as **GradientActiveCaption**.

5. Next, add **clear button** and **'.'** Button. Then place them on either side of 0. Change its **name** to **bc** and **bdot** and **text** property to **C** and **'.'** respectively. Similarly create **add, sub, mul** and **div** buttons as shown below:

Change the name property as follows.

- a. + name is badd
- b. - name is bsub
- c. * name is bmul
- d. / name is bdiv
- e. = name is bequal

6. Add the functionalities for each button to make the calculator work properly

The final output should look similar to the one given below.

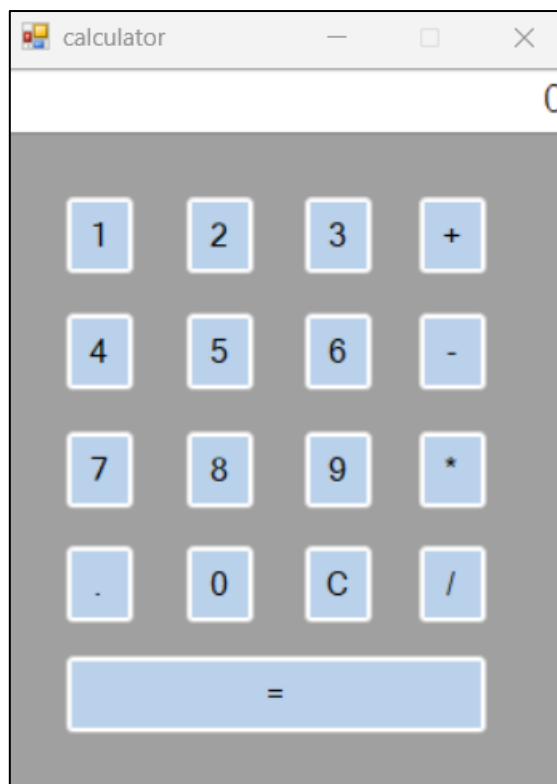


Figure 3: Final Calculator App